

Conformal Mappings and the Schwarz-Christoffel Transformation

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by
Rebecca Heinen
Dr. Loukas Grafakos, Thesis Supervisor

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The undersigned, appointed by the Dean of the Graduate School, have examined the thesis entitled

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presented by Rebecca Heinen, a candidate for the degree of Master of Arts, and hereby certify that in their opinion it is worthy of acceptance.

Professor Loukas Grafakos

Professor Nakhlé Asmar

Professor David Retzloff

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Rebecca Heinen

Dr. Loukas Grafakos, Thesis Supervisor

ABSTRACT

Let Ω be an open and connected set in the complex plane. A mapping $f : \Omega \rightarrow \mathbb{C}$ is said to be conformal at a point z_0 if it preserves angles and orientation between curves intersecting at z_0 . We discuss tangent lines and their relationship to the conformal property. This property is useful for determining the points at which a function is conformal. We provide several examples and a summary of boundary behavior properties.

We discuss two types of conformal mappings, linear fractional transformations and the Schwarz-Christoffel transformation. Linear fractional transformations map lines and circles to either lines or circles; meanwhile, the Schwarz-Christoffel transformation maps the upper half-plane onto a polygon P . We prove several properties of these mappings and demonstrate their importance with the use of examples.

These results are useful for solving boundary value problems on irregular regions. If we map such a region onto another region whose solution is known, we are able to then obtain a solution on the first region. Important physical applications of this thesis include determination of isotherms, streamlines, and air gap reluctance for magnetic circuits.

Chapter 1

Introduction to Conformal Mappings

Our goal in this chapter is to introduce basic properties of mappings given by analytic functions. We will begin with a brief recall of tangent lines.

Basic Properties

Let γ be a smooth path parametrized by $z(t) = x(t) + iy(t)$ with $a \leq t \leq b$. To ensure the tangent exists, also suppose that $z'(t) = x'(t) + iy'(t) \neq 0$ for all $a \leq t \leq b$. We see in Figure ?? that the tangent line to the curve at $z_0 = z(t)$ has the direction of $z'(t)$. Furthermore, we can determine the direction of $z'(t)$ by $\arg(z'(t))$.

Figure 1.1: The direction of the tangent line at $z(t)$ is given by $\arg(z'(t))$.

Figure 1.2: The curves γ_1 and γ_2 intersect at angle α .

We let z_0 be a point in the z -plane and consider two smooth paths γ_1 and γ_2 that intersect at z_0 . Let L_1 and L_2 be the tangent lines to γ_1 and γ_2 , respectively, at z_0 . The curves γ_1 and γ_2 intersect at angle α at z_0 if L_1 and L_2 intersect at angle α at z_0 . This is demonstrated in Figure ??.

Figure 1.3: A linear mapping $f(z) = az + b$ ($a \neq 0$) rotates by an angle $\arg(a)$, dilates by a factor $|a|$, and translates by b . In particular, it preserves angles between curves.

To begin our discussion on conformal mappings, we will first explore a simple example. Consider the linear mapping $f(z) = az + b$ where $a, b \in \mathbb{C}$ and $a \neq 0$. Under the usual interpretation, f takes points in the z -plane and maps them to points in w -plane. The map f rotates points in z -plane by the fixed angle $\arg(a)$, dilates points by a factor of $|a|$, and translates points by b . If our smooth paths γ_1 and γ_2 intersect at angle α at z_0 in the z -plane, then the images of these curves under f are two paths in the w -plane that intersect at $w_0 = f(z_0)$. It is important to note that the angle at which the images of these curves intersect in the w -plane is also α because each curve is rotated by the same angle $\arg(a)$ under f . Moreover, f preserves the orientation of γ_1 with respect to γ_2 .

Definition 1.1. Let Ω be a region, an open and connected set. A mapping $f : \Omega \rightarrow \mathbb{C}$ is said to be conformal at a point z_0 if it preserves angles and orientation between curves intersecting at z_0 .

To motivate our first theorem, we make the following observation. Suppose f is analytic at z_0 and $f'(z_0) \neq 0$. It follows that f is approximately linear in a neighborhood of z_0 . That is,

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \varepsilon(z)(z - z_0),$$

where $\varepsilon(z) \rightarrow 0$ as $z \rightarrow z_0$. We observe that $f(z) \approx f'(z_0)z + b$ where $b = -f'(z_0)z_0 + f(z_0)$ is a constant near z_0 . We see that f is approximately linear, which suggests that f is conformal at z_0 . The map f rotates by angle $\theta = \arg(f'(z_0))$, scales by a factor of $|f'(z_0)|$, and translates by $b = -f'(z_0)z_0 + f(z_0)$.

Example 1.2. We determine the angle of rotation $\alpha = \arg(f'(z))$ and the dilation factor $|f'(z)|$ of the given mappings at the indicated points.

- (a) $\text{Log}(z)$ at $z = 1, i, -i$
- (b) $\frac{1+e^z}{1-e^z}$ at $z = 1, i\pi, \ln(3) + 2i$

Solution.

(a) Notice that $f(z) = \text{Log}(z)$ is analytic on its domain, which is the complex plane with the principal branch cut. Note $f'(z) = \frac{1}{z} \neq 0$ for all $z \in \mathbb{C}$. When $z = 1$, $\alpha = \arg(f'(1)) = \arg(1) = 0 + 2\pi k = 2\pi k$ where $k \in \mathbb{Z}$ and we have a dilation factor of $|f'(1)| = |1| = 1$. When $z = i$, $\alpha = \arg(f'(i)) = \arg(f'(-i)) = -\frac{\pi}{2} + 2\pi k$ for $k \in \mathbb{Z}$ and we have a dilation factor of $|f'(i)| = |-i| = 1$. When $z = -i$, $\alpha = \arg(f'(-i)) = \arg(i) = \frac{\pi}{2} + 2\pi k$ for $k \in \mathbb{Z}$ and we have a dilation factor of $|f'(-i)| = |i| = 1$.

(b) Notice that $f(z) = \frac{1+e^z}{1-e^z}$ is analytic for $\mathbb{C} \setminus \{2k\pi i, k \in \mathbb{Z}\}$ and $f'(z) = \frac{2e^z}{(1-e^z)^2} \neq 0$. When $z = 1$, $\alpha = \arg(f'(1)) = \arg\left(\frac{2e}{(1-e)^2}\right) = 0 + 2\pi k = 2\pi k$, where $k \in \mathbb{Z}$, and we have a dilation factor of $|f'(1)| = \left|\frac{2e}{(1-e)^2}\right| = \frac{2e}{(1-e)^2}$. When $z = i\pi$, $\alpha = \arg(f'(i\pi)) = \arg\left(\frac{2e^{i\pi}}{1-e^{i\pi}}\right) = \arg(-1) = \pi + 2\pi k$ where $k \in \mathbb{Z}$. We have a dilation factor of $|f'(i\pi)| = |-1| = 1$. When $z = \ln(3) + 2i$, $\alpha = \arg(f'(\ln(3) + 2i))$. To calculate $f'(\ln(3) + 2i)$,

$$\begin{aligned} f'(\ln(3) + 2i) &= \frac{6e^{2i}}{(1 - 3e^{2i})^2} \\ &= \frac{6e^{2i}}{(1 - 3e^{2i})^2} \frac{(1 - 3e^{-2i})^2}{(1 - 3e^{-2i})^2} \\ &= \frac{6e^{2i}(1 - 6e^{-2i} + 9e^{-4i})}{(10 - 6\cos(2))^2}. \end{aligned}$$

Now,

$$\arg\left(\frac{6e^{2i}(1 - 6e^{-2i} + 9e^{-4i})}{(10 - 6\cos(2))^2}\right) = \arg\left(\frac{6}{(10 - 6\cos(2))^2}\right) +$$

$$\begin{aligned}
& \arg(e^{2i}(1 - 6e^{-2i} + 9e^{-4i})) \\
&= 0 + 2\pi l + \arg(\cos(2) + i \sin(2) - 6 + \\
& \quad 9 \cos(2) - 9i \sin(2)) \\
&= \arg(10 \cos(2) - 6 - 8i \sin(2)) + 2\pi l \\
&= \arctan\left(\frac{-4 \sin(2)}{5 \cos(2) - 3}\right) + 2\pi k,
\end{aligned}$$

where $l, k \in \mathbb{Z}$. Thus, $\alpha = \arctan\left(\frac{-4 \sin(2)}{5 \cos(2) - 3}\right) + 2\pi k$, $k \in \mathbb{Z}$. We have a dilation factor of

$$\begin{aligned}
|f'(\ln(3) + 2)| &= \left| \frac{6e^{2i}(1 - 6e^{-2i} + 9e^{-4i})}{(10 - 6 \cos(2))^2} \right| \\
&= \frac{6\sqrt{100 \cos^2(2) - 120 \cos(2) + 36 + 64 \sin^2(2)}}{(10 - 6 \cos(2))^2}.
\end{aligned}$$

Theorem 1.3 (*Conformal Property*). *Suppose that f is analytic in a neighborhood of a point $z_0 \in \mathbb{C}$ and suppose that $f'(z_0) \neq 0$. Then f is conformal at z_0 .*

Proof. Let γ be any smooth path through z_0 , parametrized by $\gamma(t) = x(t) + iy(t)$ with $\gamma(t_0) = z_0$ and $\gamma'(t_0) \neq 0$. The image of γ under f is a path parametrized by $f(\gamma(t))$ that passes through $w_0 = f(z_0)$ in the w -plane. Since $\gamma'(t_0) \neq 0$, the direction of the tangent line to γ at z_0 is $\arg(\gamma'(t_0))$. By the chain rule,

$$(f \circ \gamma)'(t_0) = f'(\gamma(t_0))\gamma'(t_0) \neq 0$$

because we assumed $f'(z_0) = f'(\gamma(t_0)) \neq 0$ and $\gamma'(t_0) \neq 0$. The direction the tangent line to $f(\gamma(t))$ at $w_0 = f(z_0)$ is

$$\arg((f \circ \gamma)'(t_0)) = \arg(f'(\gamma(t_0))\gamma'(t_0)) = \arg(f'(\gamma(t_0))) + \arg(\gamma'(t_0)).$$

We assumed that $\gamma(t_0) = z_0$, so this equation yields

$$\arg((f \circ \gamma)'(t_0)) = \arg(f'(z_0)) + \arg(\gamma'(t_0)).$$

Let γ_1 and γ_2 be two curves with $\gamma_1(t_0) = \gamma_2(t_0) = z_0$. We have the following two equations:

$$\arg((f \circ \gamma_1)'(t_0)) = \arg(f'(z_0)) + \arg(\gamma_1'(t_0))$$

and

$$\arg((f \circ \gamma_2)'(t_0)) = \arg(f'(z_0)) + \arg(\gamma_2'(t_0)).$$

Subtracting these two equations gives us

$$\arg((f \circ \gamma_1)'(t_0)) - \arg((f \circ \gamma_2)'(t_0)) = \arg(\gamma_1'(t_0)) - \arg(\gamma_2'(t_0)).$$

We see that the oriented angle of the intersection between any two tangent lines of curves intersecting at z_0 is the same as that for the images of these curves under the mapping f . Furthermore, f is a conformal mapping by definition. ■

The following examples demonstrate the importance of Theorem ?? and its applications.

Example 1.4. We determine where the given mappings are conformal.

$$(a) \ f(z) = \frac{\sin(z)}{e^z}$$

$$(b) \ f(z) = \frac{e^z + 1}{e^z - 1}$$

$$(c) \ f(z) = \frac{z + 1}{z + i} + \frac{2}{z}$$

Solution.

(a) Notice for $f(z) = \frac{\sin(z)}{e^z}$ that $f'(z) = \frac{\cos(z) - \sin(z)}{e^z}$. The fact that $\sin(z)$ and e^z are entire and $e^z \neq 0$ implies $f(z) = \frac{\sin(z)}{e^z}$ is also entire. Notice that $f' = 0$ when $\cos(z) - \sin(z) = 0$. This occurs when $z = \frac{\pi}{4} + k\pi$ for $k \in \mathbb{Z}$. Thus, f is conformal for all $z \neq \frac{\pi}{4} + k\pi$ where $k \in \mathbb{Z}$ by Theorem ??.

(b) Notice for $f(z) = \frac{e^z+1}{e^z-1}$ that $f'(z) = \frac{-2e^z}{(e^z-1)^2}$. The fact that $e^z + 1$ and $e^z - 1$ are entire and $e^z - 1 = 0$ for $z = 2k\pi i$ where $k \in \mathbb{Z}$ implies that $f(z) = \frac{e^z+1}{e^z-1}$ is analytic for all $z \neq 2k\pi i$ for $k \in \mathbb{Z}$. Notice that $f' \neq 0$ because $e^z \neq 0$. Thus, f is conformal for all $z \neq 2k\pi i$ where $k \in \mathbb{Z}$ by Theorem ??.

(c) Notice for $f(z) = \frac{z+1}{z+i} + \frac{2}{z}$ that $f'(z) = \frac{i-1}{(z+i)^2} - \frac{2}{z^2}$. We know that $\frac{z+1}{z+i}$ is analytic for $\mathbb{C} \setminus \{-i\}$ and $\frac{2}{z}$ is analytic for $\mathbb{C} \setminus \{0\}$ and so f is analytic for $\mathbb{C} \setminus \{-i, 0\}$. To determine when $f' = 0$, it remains find z such that $\frac{i-1}{(z+i)^2} = \frac{2}{z^2}$. In other words, we will solve $(i-3)z^2 - 4iz + 2 = 0$. Using the quadratic formula, we have that

$$\begin{aligned} z &= \frac{4i \pm \sqrt{(-4i)^2 - 4(i-3)(2)}}{2(i-3)} \\ &= \frac{4i \pm \sqrt{8-8i}}{2i-6}. \end{aligned}$$

Next, we calculate $\sqrt{8-8i}$. We know that $r = 8\sqrt{2}$ and $\text{Arg}(8-8i) = -\frac{\pi}{4}$. Thus, our two roots are

$$\begin{aligned} w_0 &= 2\sqrt{2}\sqrt[4]{2} \left(\cos\left(-\frac{\pi}{8}\right) + i \sin\left(-\frac{\pi}{8}\right) \right) \\ w_1 &= 2\sqrt{2}\sqrt[4]{2} \left(\cos\left(\frac{7\pi}{8}\right) + i \sin\left(\frac{7\pi}{8}\right) \right) \\ &= 2\sqrt{2}\sqrt[4]{2} \left(\cos\left(-\frac{\pi}{8} + \pi\right) + i \sin\left(-\frac{\pi}{8} + \pi\right) \right) \\ &= -w_0. \end{aligned}$$

Therefore, $z = \frac{4i+w_0}{2i-6}$ or $z = \frac{4i+w_1}{2i-6}$ where w_0 and w_1 are as above. We conclude then that f is conformal for all $z \neq -i, 0, \frac{4i+w_0}{2i-6}, \frac{4i+w_1}{2i-6}$ by Theorem ??.

Example 1.5. We consider the orthogonal lines $x = a$ and $y = b$, where $a, b \in \mathbb{R}$. We determine their images under the given mappings. We also determine the values of a and b such that the images are orthogonal at the point of intersection $f(a + ib)$ and verify by computing the angle between the image curves at the point $f(a + ib)$.

(a) $f(z) = (1 + i)z$

(b) $f(z) = \frac{z+1}{z-1}$

Solution.

(a) Notice that $f(z) = (1 + i)z$ is analytic and $f'(z) = 1 + i \neq 0$. This implies that f is conformal for all $\mathbb{C}$. The image of $x = a$ ($z = a + iy$ for $y \in \mathbb{R}$) is $f(a + iy) = (1 + i)(a + iy) = (a - y) + i(a + y)$. The image of $y = b$ ($z = x + ib$ for $x \in \mathbb{R}$) is $f(x + ib) = (1 + i)(x + ib) = (x - b) + i(x + b)$. When $x = a$, we have $(a - y, a + y)$ by taking the real and imaginary parts of $f(a + iy)$. Likewise for $y = b$ we have $(x - b, x + b)$. The tangent vectors of these curves are $(-1, 1)$ and $(1, 1)$. Notice that $(-1, 1) \cdot (1, 1) = 0$ and so the images of these two lines are indeed orthogonal.

(b) Notice that $f(z) = \frac{z+1}{z-1}$ is analytic for all $z \in \mathbb{C} \setminus \{1\}$ and $f'(z) = \frac{-2}{(z-1)^2} \neq 0$. This implies that f is conformal for all $z \neq 1$. Therefore, the images of $x = a$ and $y = b$ should be orthogonal at every value except $x = 1$. Notice that

$$f(z) = \frac{z+1}{z-1} = \frac{x^2 + y^2 - 1}{(x-1)^2 + y^2} + i \left(\frac{-2y}{(x-1)^2 + y^2} \right).$$

When $x = a$, we have the vector $\left(\frac{a^2 + y^2 - 1}{(a-1)^2 + y^2}, \frac{-2y}{(a-1)^2 + y^2} \right)$ and so the tangent vector at the point (a, b) is $\left(\frac{4b - 4ab}{((a-1)^2 + b^2)^2}, \frac{-2a^2 + 4a - 2 + 2b^2}{((a-1)^2 + b^2)^2} \right)$. When $y = b$, we have the vector $\left(\frac{x^2 + b^2 - 1}{(x-1)^2 + b^2}, \frac{-2b}{(x-1)^2 + b^2} \right)$ and so the tangent vector at the point (a, b) is $\left(\frac{-2a^2 + 4a + 2b^2 - 2}{((a-1)^2 + b^2)^2}, \frac{4ba - 4b}{((a-1)^2 + b^2)^2} \right)$. Notice that dot product between these two tangent vectors is zero so the images of these two curves are indeed orthogonal.

Boundary Behavior

When solving boundary value problems, we would like to know if the boundary of a region Ω is mapped onto the boundary of $f[\Omega]$ under a map f . Conformal mappings are useful for solving boundary value problems on irregular regions by mapping them onto regions for which the particular boundary value problem can be easily solved. However, under what conditions will the boundary of Ω be mapped onto the boundary of $f[\Omega]$? The following example illustrates that the property of being conformal alone is not sufficient.

Example 1.6. Consider $f(z) = z^2$ and $\Omega = \{z = re^{i\theta} : \frac{1}{2} < r < 1, -\pi < \theta < \pi\}$. Here f is analytic and $f'(z) = 2z \neq 0$ for all z in Ω . By Theorem ??, f is conformal for all $z \in \Omega$. Using DeMoivre's identity, we see that

$$f[\Omega] = \{w = re^{i\theta} : \frac{1}{4} < r < 1, 0 \leq \theta \leq 2\pi\}$$

Figure 1.4: The boundary of Ω is not mapped to boundary of $f[\Omega]$ under f .

Although $f(z) = z^2$ is conformal in Ω , the interval $(-1, -\frac{1}{2})$ on the boundary of Ω is mapped to the interval $(\frac{1}{4}, 1)$ in the interior of $f[\Omega]$. We conclude that the boundary of Ω is not mapped onto the boundary of $f[\Omega]$ under the map f .

The condition that $f' \neq 0$ for all $z \in \Omega$ ensures that f is one-to-one locally at $z \in \Omega$ by the inverse function theorem [?]. However, the function may fail to be one-to-one on the entire region Ω . This is exactly what happened in Example ??, so this leads us to believe that the property of being one-to-one is necessary for a function f to map boundary points to boundary points. We have the result that if

f is analytic and nonconstant on a region Ω , then $f[\Omega]$ is a region [?]. This implies all the points in Ω are mapped to the open connected set $f[\Omega]$. It is possible that f is not defined on the boundary of Ω , so we will introduce the following definitions to describe how f maps the boundary of Ω .

Definition 1.7. Let f be a mapping from a region Ω to the complex plane.

- (i) We say that f **maps the boundary of Ω to the boundary of $f[\Omega]$** if the following condition holds: If $\{z_n\}_{n=1}^{\infty}$ is any sequence in Ω converging to α_0 on the boundary of Ω and β is any point in $f[\Omega]$, then $\{f(z_n)\}_{n=1}^{\infty}$ does not converge to β . So if $\{f(z_n)\}_{n=1}^{\infty}$ converges, it must converge to a boundary point of $f[\Omega]$ or to infinity. If the region Ω is unbounded, we allow α_0 to be ∞ .
- (ii) We say that f **maps the boundary of Ω onto the boundary of $f[\Omega]$** if f maps the boundary of Ω to the boundary of $f[\Omega]$ and, additionally, for every point w_0 on the boundary of $f[\Omega]$ there is a sequence $\{z_n\}_{n=1}^{\infty}$ in Ω with $z_n \rightarrow \alpha_0$, α_0 a boundary point of Ω , such that $f(z_n) \rightarrow w_0$.

There are cases where $\{f(z_n)\}_{n=1}^{\infty}$ does not converge, but $\{z_n\}_{n=1}^{\infty}$ converges to a point on the boundary of a region. We will see this in the next example.

Example 1.8. Let Ω be the slit plane, $\mathbb{C} \setminus \{z : z \leq 0\}$, and $f(z) = \text{Log}(z)$. We want to show that $z_n = -1 + i\frac{(-1)^n}{n}$ ($n = 1, 2, \dots$) converges to -1 on the boundary of Ω . To show this, notice that for $n = 2k$, then $z_n = z_{2k} = -1 + \frac{i}{2k}$, which converges to -1 as $k \rightarrow \infty$. Also, notice that for $n = 2k + 1$, then $z_n = z_{2k+1} = -1 - \frac{i}{2k+1}$, which converges to -1 as $k \rightarrow \infty$. Thus, we conclude that $z_n \rightarrow -1$ as $n \rightarrow \infty$ where -1 is on the boundary of Ω . Now, we will show that $\{f(z_n)\}_{n=1}^{\infty}$ does not converge. Notice

that

$$\begin{aligned}\operatorname{Log}(z_n) &= \ln \left| -1 + i \frac{(-1)^n}{n} \right| + i \operatorname{Arg} \left(-1 + i \frac{(-1)^n}{n} \right) \\ &= \ln \left| \sqrt{1 + \frac{1}{n^2}} \right| + i \operatorname{Arg} \left(-1 + i \frac{(-1)^n}{n} \right).\end{aligned}$$

When $n = 2k$, $z_{2k} \rightarrow -1$ from the upper-plane implies that $\operatorname{Arg}(z_{2k}) \rightarrow \pi$ as $k \rightarrow \infty$.

Thus, $\operatorname{Log}(z_{2k}) \rightarrow \ln(1) + i(\pi) = i\pi$ as $k \rightarrow \infty$. When $n = 2k + 1$, $z_{2k+1} \rightarrow -1$ from the lower-plane implies that $\operatorname{Arg}(z_{2k+1}) \rightarrow -\pi$ as $k \rightarrow \infty$. Thus, $\operatorname{Log}(z_{2k+1}) \rightarrow \ln(1) + i(-\pi) = -i\pi$ as $k \rightarrow \infty$. Since $\operatorname{Log}(z_{2k}) \rightarrow i\pi$ and $\operatorname{Log}(z_{2k+1}) \rightarrow -i\pi$ as $k \rightarrow \infty$, we conclude that $\{\operatorname{Log}(z_n)\}_{n=1}^{\infty}$ does not converge.

Theorem 1.9. *Suppose that f is analytic and one-to-one on a region Ω . Then, f maps the boundary of Ω onto the boundary of $f[\Omega]$.*

Proof. Suppose that f is analytic and one-to-one on a region Ω . First, we will show that f maps the boundary of Ω to the boundary of $f[\Omega]$. The fact that f is one-to-one and nonconstant implies that $f[\Omega]$ is a region. Let β be a point in $f[\Omega]$ and suppose, to obtain a contradiction, that $\{z_n\}_{n=1}^{\infty}$ is a sequence in Ω converging to a boundary point α_0 of Ω with $f(z_n) \rightarrow \beta$. Notice that f^{-1} exists and is analytic as f is analytic and one-to-one. Moreover, f^{-1} is continuous and so

$$\lim_{n \rightarrow \infty} z_n = \lim_{n \rightarrow \infty} f^{-1}(f(z_n)) = f^{-1}(\lim_{n \rightarrow \infty} f(z_n)) = f^{-1}(\beta).$$

Note that β being a point in $f[\Omega]$ implies that $f^{-1}(\beta)$ is a point in Ω . Therefore, we see that $\{z_n\}_{n=1}^{\infty}$ converges to an interior point of Ω , a contradiction. We conclude that if $\{z_n\}_{n=1}^{\infty}$ converges to a point α_0 on the boundary of Ω , then $\{f(z_n)\}_{n=1}^{\infty}$ cannot converge to β . Second, we will show that f maps the boundary of Ω onto the boundary

of $f[\Omega]$. To do so, let w_0 be a point on the boundary of $f[\Omega]$. There exist points $w_n = f(z_n)$ such that z_n is in Ω and $w_n \rightarrow w_0$. Two cases arise. First, suppose $\{z_n\}_{n=1}^\infty$ is an unbounded sequence. There exists a subsequence $\{z_{n_j}\}_{j=1}^\infty$ that tends to ∞ , which is a point on the boundary of Ω , if Ω is an unbounded region. Moreover, $f(z_{n_j}) \rightarrow w_0$. Second, we consider the case in which $\{z_n\}_{n=1}^\infty$ is a bounded sequence. Using Bolzano-Weierstrass, we can find a subsequence $\{z_{n_j}\}_{j=1}^\infty$ that converges to a point α_0 in $\overline{\Omega}$, the closure of Ω . If α_0 were in Ω , then we would have $f(z_{n_j}) \rightarrow f(\alpha_0)$ as f is continuous. We also know that $f(z_{n_j}) \rightarrow w_0$, so we would have $f(\alpha_0) = w_0$. This produces a contradiction as $f(\alpha_0)$ would lie in $f[\Omega]$ and w_0 lies on the boundary of $f[\Omega]$. Thus, α_0 lies on the boundary of Ω . ■

Corollary 1.10. *Suppose that f is analytic and one-to-one in a region Ω . Suppose that f extends continuously to Ω . Then for any point α of the boundary of Ω the point $f(\alpha)$ lies on the boundary of $f[\Omega]$.*

Proof. Let $\{z_n\}_{n=1}^\infty$ be a sequence in Ω with $z_n \rightarrow \alpha$ where α is on the boundary of Ω . By the continuity of f , we have that $f(z_n) \rightarrow f(\alpha)$. We know that $f(z_n)$ does not converge to an interior point of Ω by Theorem ??, so $f(\alpha)$ must be a boundary point of $f[\Omega]$. ■

These results about boundary behavior are particularly important for solving Dirichlet problems, but we may also use these results when constructing mappings from regions to regions. We will now discuss particular types of conformal mappings, linear fractional transformations and Schwarz-Christoffel transformations, in the following two chapters.

Chapter 2

Linear Fractional Transformations

We are mostly interested in solving boundary value problems, such as the Dirichlet problem, on the unit disk. Given a region Ω that is not the unit disk we will want to find a conformal, one-to-one map that maps Ω onto the unit disk. It turns out that the best and most convenient mappings are linear fractional transformations:

$$\phi(z) = \frac{az + b}{cz + d}, \quad (ad \neq bc) \quad (2.1)$$

The condition $ad \neq bc$ ensures that $\phi'(z) = \frac{ad - bc}{(cz + d)^2} \neq 0$ for all $z \in \mathbb{C}$, which implies ϕ is not a constant function. If $c = 0$, then ϕ is a linear function, which is entire. If $c \neq 0$, then ϕ is analytic for all $z \neq -\frac{d}{c}$ and has a simple pole at $z = -\frac{d}{c}$.

Proposition 2.1. *Let ϕ be the linear fractional transformation defined in (2.1) where $a, b, c, d \in \mathbb{C}$ with $c \neq 0$ and $ad \neq bc$. Then,*

(i) *ϕ has a pole at $z = -\frac{d}{c}$ and is one-to-one and conformal on $\mathbb{C} \setminus \{-\frac{d}{c}\}$ and its range is $\mathbb{C} \setminus \{\frac{a}{c}\}$.*

(ii) *The inverse of ϕ is the linear fractional transformation from $\mathbb{C} \setminus \{\frac{a}{c}\}$ onto $\mathbb{C} \setminus \{-\frac{d}{c}\}$ given by*

$$\psi(w) = \frac{dw - b}{-cw + a}. \quad (2.2)$$

Proof. (i) Let $\phi(z) = \frac{az+b}{cz+d}$ with $a, b, c, d \in \mathbb{C}$, $c \neq 0$, and $ad \neq bc$. Notice that ϕ has a simple pole at $z = -\frac{d}{c}$. Furthermore, $\phi'(z) = \frac{ad-bc}{(cz+d)^2} \neq 0$ for all $z \in \mathbb{C} \setminus \{-\frac{d}{c}\}$. By Theorem ??, ϕ is conformal at $z \neq -\frac{d}{c}$. To show that ϕ is one-to-one, let $z_1, z_2 \in \mathbb{C} \setminus \{-\frac{d}{c}\}$ and suppose $\phi(z_1) = \phi(z_2)$. Using the fact that $\phi(z_1) = \phi(z_2)$ and multiplying by $(cz_1 + d)(cz_2 + d)$, we obtain

$$(ad - bc)(z_1 - z_2) = 0.$$

We assumed that $ad - bc \neq 0$, so it must be that $z_1 - z_2 = 0$. Hence, $z_1 = z_2$ and so ϕ is one-to-one. To determine the range of ϕ , let $z^* \in \mathbb{C}$. We want to find a $z \in \mathbb{C}$ such that $\phi(z) = z^*$. Here $z = \frac{-b+z^*d}{a-z^*c}$ which is not defined for $z^* = \frac{a}{c}$. Furthermore, the range of ϕ is $\mathbb{C} \setminus \{\frac{a}{c}\}$. (ii) Notice that $\phi : \mathbb{C} \setminus \{-\frac{d}{c}\} \rightarrow \mathbb{C} \setminus \{\frac{a}{c}\}$ is bijective and so ϕ has an inverse. Namely, it is defined by

$$\psi(w) = \frac{dw - b}{a - wc}.$$

Therefore, we conclude that (i) and (ii) hold. ■

We now observe that every such linear fractional transformation with $c \neq 0$ is the composition of a linear mapping $w_1 = cz + d$, $w_2 = \frac{1}{w_1}$, and a linear mapping $w = \frac{a}{c} + (b - \frac{ad}{c})w_2$. Linear mappings map lines to lines and circles to circles. This is quickly verified by the fact that a linear mapping $f(z) = az + b$ rotates by an angle $\arg(a)$, dilates by a factor of $|a|$, and translates by b units. On the other hand, the inversion map $f(z) = \frac{1}{z}$ maps lines and circles to either lines or circles. This is more difficult to see, but we will prove the following results about the inversion map.

Lemma 2.2. *Any circle of the form $(x - x_0)^2 + (y - y_0)^2 = r^2$ for $r > 0$ and any line*

in the plane can be written in the form

$$A(x^2 + y^2) + Bx + Cy + D = 0,$$

where $B^2 + C^2 - 4AD > 0$.

Proof. If we have a circle of the form $(x - x_0)^2 + (y - y_0)^2 = r^2$ for $r > 0$, we can obtain $(x^2 + y^2) - 2x_0x - 2y_0y + (x_0^2 + y_0^2 - r^2) = 0$ by expanding the initial terms and grouping the appropriate terms. We see that $A = 1$, $B = -2x_0$, $C = -2y_0$, and $D = x_0^2 + y_0^2 - r^2$. We know that a line is of the form $y = mx + b$, which implies $y - mx - b = 0$. We see that $A = 0$, $B = -m$, $C = 1$, and $D = -b$. ■

Lemma 2.3. *Any set of points satisfying $A(x^2 + y^2) + Bx + Cy + D = 0$ with $B^2 + C^2 - 4AD > 0$ is either a circle or a line depending on whether $A = 0$ or $A \neq 0$. Furthermore, such circles and lines pass through the origin if and only if $D = 0$.*

Proof. If $A = 0$, then we have $Bx + Cy + D = 0$ where $B^2 + C^2 > 0$. The inequality $B^2 + C^2 > 0$ ensure that both B and C cannot be zero at the same time. Notice that $Bx + Cy + D = 0$ is a line. When $A \neq 0$, then we have $A(x^2 + y^2) + Bx + Cy + D = 0$. Completing the square yields $(x + \frac{B}{2A})^2 + (y + \frac{C}{2A})^2 = \frac{B^2 + C^2 - 4AD}{4A^2}$. The fact that $B^2 + C^2 - 4AD > 0$ implies we have a circle with center $(-\frac{B}{2A}, -\frac{C}{2A})$ and radius $r = \frac{\sqrt{B^2 + C^2 - 4AD}}{2A}$. A line is of the form $Bx + Cy + D = 0$ where $B^2 + C^2 > 0$. If $D = 0$, then we simply have $Bx + Cy = 0$, which is a line passing through the origin. If the line passes through the origin (i.e. $(x, y) = (0, 0)$), we have $D = 0$. A circle is of the form $(x + \frac{B}{2A})^2 + (y + \frac{C}{2A})^2 = \frac{B^2 + C^2 - 4AD}{4A^2}$. If $D = 0$, this reduces to a circle of the form $(x + \frac{B}{2A})^2 + (y + \frac{C}{2A})^2 = \frac{B^2 + C^2}{4A^2}$. Note that the distance between the center $(-\frac{B}{2A}, -\frac{C}{2A})$ and the origin is $\frac{\sqrt{B^2 + C^2}}{2A}$, which is precisely the radius of this

circle. Therefore, the circle must pass through the origin. On the other hand, if $x = y = 0$, we have that $\frac{B^2+C^2}{4A^2} = \frac{B^2+C^2-4AD}{4A^2}$. This implies that $-4AD = 0$ and so $D = 0$ as $A \neq 0$. ■

Proposition 2.4. *Let S be a set of points (x, y) satisfying $A(x^2+y^2)+Bx+Cy+D = 0$ where $B^2 + C^2 - 4AD > 0$. Then, under the map $f(z) = \frac{1}{z}$, points (u, v) in $f[S]$ satisfy an equation of the same form. Moreover, lines and circles are mapped to lines and circles, with the exception of the origin.*

Proof. Assume S is a set of points (x, y) satisfying $A(x^2 + y^2) + Bx + Cy + D = 0$ where $B^2 + C^2 - 4AD > 0$. By Lemma ??, S is either a line or circle. For any point $z \in S$, we have

$$\frac{1}{z} = \frac{1}{x + iy} = \frac{x - iy}{x^2 + y^2}.$$

We conclude that

$$f[S] = \left\{ u + iv : u = \frac{x}{x^2 + y^2}, v = \frac{-y}{x^2 + y^2} \right\}$$

with x and y satisfying $A(x^2 + y^2) + Bx + Cy + D = 0$ with $B^2 + C^2 - 4AD > 0$. We want to prove that u and v satisfy a similar identity with possibly different values of the coefficients. Notice that

$$\begin{aligned} E(u^2 + v^2) + Fu + Gv + H &= E \frac{x^2 + y^2}{(x^2 + y^2)^2} + \frac{Fx}{x^2 + y^2} - \frac{Gy}{x^2 + y^2} + H \\ &= \frac{E}{x^2 + y^2} + \frac{Fx}{x^2 + y^2} - \frac{Gy}{x^2 + y^2} + \frac{H(x^2 + y^2)}{x^2 + y^2} \\ &= \frac{1}{x^2 + y^2} (H(x^2 + y^2) + Fx - Gy + E). \end{aligned}$$

If we choose $H = A$, $F = B$, $G = -C$, and $E = D$, then, for all points in $S \setminus \{(0, 0)\}$, the last expression is equal to 0 by assumption. We conclude that (u, v) satisfy

$$D(u^2 + v^2) + Bu - Cv + A = 0$$

with $B^2 + C^2 - 4AD > 0$. By Lemma ??, we conclude that $f[S]$ is either a line or circle. ■

Because linear fractional transformations are a composition of two linear mappings and the inversion map, linear fractional transformations will map lines to either lines or circles and circles to either lines or circles. We illustrate this in the next example.

Example 2.5. Let L_1 denote the line through z_1 and z_2 and L_2 the line through z_2 and z_3 . We describe the images by ϕ of L_1 and L_2 .

$$(a) \quad \phi(z) = i \frac{1-z}{1+z}, z_1 = 1, z_2 = 0, z_3 = i$$

$$(b) \quad \phi(z) = \frac{1+i-2z}{i-iz}, z_1 = i, z_2 = 1, z_3 = -i$$

Solution.

(a) Let L_1 be the line through $z_1 = 1$ and $z_2 = 0$ and L_2 be the line through $z_2 = 0$ and $z_3 = i$. Notice that the L_1 is perpendicular to L_2 , so we would expect $\phi[L_1]$ to be perpendicular to $\phi[L_2]$. First, we consider $\phi[L_1]$. Notice that the point $z = -1$ is on L_1 and $\phi(-1) = \infty$. Thus, we know that $\phi[L_1]$ is a line. Since $\phi(1) = 0$ and $\phi(0) = i$, we conclude that $\phi[L_1]$ is the imaginary axis in the w -plane. Since $\phi[L_1]$ is the imaginary axis and $\phi[L_2]$ must go through $w_2 = \phi(0) = i$ and $w_3 = \phi(i) = 1$, we would expect that $\phi[L_2]$ is a circle. In addition to the points w_2 and w_3 , we have that $\phi(-i) = -1$. This tells us that $\phi[L_2]$ is a circle through the points $w_2 = i$, $w_3 = 1$, and $w_4 = -1$.

(b) Let L_1 denote the line through the points $z_1 = i$ and $z_2 = 1$ and L_2 denote the line through the points $z_2 = 1$ and $z_3 = -i$. Notice that $\phi[L_1]$ will go through

$w_1 = -i$ and $w_2 = \infty$ and so we conclude that $\phi[L_1]$ is a line through the point $w_1 = -i$. To determine the line, we use the point $z = \frac{1+i}{2}$ as a second point. Notice that $w_4 = \phi\left(\frac{1+i}{2}\right) = 0$. Therefore, we conclude that $\phi[L_1]$ is the imaginary axis in the w -plane. Next, $\phi[L_2]$ will go through $w_2 = \infty$ and $w_3 = 1 - 2i$, which tells us that $\phi[L_2]$ is a line. To determine the line, we use $z = \frac{1-i}{2}$ as a second point. Notice that $w_5 = \phi\left(\frac{1-i}{2}\right) = 2 - 2i$. Thus, $\phi[L_2]$ is a line going through the points $w_3 = 1 - 2i$ and $w_5 = 2 - 2i$.

We will now address two common mappings that are used to transfer between the unit disk and the upper half-plane.

Example 2.6. (a) We show that the linear fractional transformation

$$\phi(z) = i \frac{1 - z}{1 + z}$$

maps the unit disk onto the upper half-plane.

(b) We show that the linear fractional transformation

$$\psi(z) = \frac{i - z}{i + z}$$

maps the upper half-plane onto the unit disk.

Solution.

(a) The image of the unit circle $C_1(0)$ is either a line or a circle. Computing the images of three points on $C_1(0)$ enables us to determine whether or not $\phi[C_1(0)]$ is a line or circle in the w -plane. Notice that $\phi(1) = 0$, $\phi(i) = 1$, and $\phi(-i) = -1$. These values lie on the u -axis of the w -plane, which suggests that $\phi[C_1(0)]$ is the u -axis. The fact that ϕ is one-to-one implies that the boundary of the unit disk is

mapped onto the boundary of its image under ϕ , the u -axis, by Theorem ???. This implies that the image of the unit disk is either the upper half-plane or the lower half-plane. Observe that $\phi(0) = i$, so we conclude that the unit disk is mapped to the upper half-plane. Note that ϕ has a singularity at $z = -1$, which enables the bounded, unit disk to be mapped onto the unbounded, upper half-plane. For the point $z = -1$ on $C_1(0)$, we see that $\lim_{z \rightarrow -1} \phi(z) = \infty$. The image of $C_1(0)$ is unbounded and thus has to be a line as a circle is bounded.

Figure 2.1: $\phi(z) = i \frac{1-z}{1+z}$ maps the unit disk to the upper half-plane.

- (b) One can see that ψ is the inverse of ϕ , so ψ will take the upper half-plane to the unit circle. Alternatively, we can proceed as in part (a). Note that $\psi(0) = 1$, $\psi(1) = i$, and $\psi(-1) = -i$. Observe the images of three collinear points are not collinear. Moreover, these points all lie on the unit circle, so the image of the x -axis under ψ is $C_1(0)$. Since $\psi(i) = 0$, we conclude that ψ maps the upper half-plane onto the unit disk.

We saw in this example that $\lim_{z \rightarrow -1} \phi(z) = \infty$ and $\lim_{z \rightarrow \infty} \psi(z) = -1$. We will generally say $\phi(z_0) = \infty$ to mean the limit at the point z_0 is infinite. Similarly, if $\lim_{z \rightarrow \infty} \frac{az+b}{cz+d} = \frac{a}{c}$, we will write $\phi(\infty) = \frac{a}{c}$.

Proposition 2.7 (Composition of Mappings). *The composition of any two linear fractional transformations is another linear fractional transformation.*

Proof. Let $\phi_1(z) = \frac{a_1z + b_1}{c_1z + d_1}$ and $\phi_2(z) = \frac{a_2z + b_2}{c_2z + d_2}$. It follows that

$$\phi(z) = (\phi_2 \circ \phi_1)(z) = \frac{(a_1a_2 + b_2c_1)z + (a_2b_1 + b_2d_1)}{(a_1c_2 + c_1d_2)z + (b_1c_2 + d_1d_2)}.$$

If we let $a = a_1a_2 + b_2c_1$, $b = a_2b_1 + b_2d_1$, $c = a_1c_2 + c_1d_2$, and $d = b_1c_2 + d_1d_2$, we see that $ad - bc = (a_1d_1 - b_1c_1)(a_2d_2 - b_2c_2) \neq 0$ as $a_1d_1 - b_1c_1 \neq 0$ and $a_2d_2 - b_2c_2 \neq 0$. Thus, ϕ is a linear fractional transformation with one pole. This pole may or may not occur at the same location as the poles of either ϕ_1 or ϕ_2 . ■

Example 2.8. We find a linear fractional transformation that maps the disk $B_2(-1)$, a disk with radius 2 and center -1 , onto the right half-plane.

Solution. We construct a linear fractional transformation f that maps the disk $B_2(-1)$ onto the right half-plane through a composition of linear fractional transformations. Note that $f_1 : B_2(-1) \rightarrow B_1(0)$ defined by $f_1(z) = \frac{1}{2}(z + 1)$ is a linear mapping. This map shifts the disk $B_2(-1)$ right one unit and shrinks the disk by a factor of two. By Example ??, $f_2(w_1) = i\frac{1-w_1}{1+w_1}$ maps the unit circle onto the upper half-plane. Lastly, the map $f_3(w_2) = -iw_2$ maps the upper half-plane onto the right half-plane by rotating the upper half-plane 90° clockwise. Note that each of these maps is a linear fractional transformation and so the composition $f(z) = f_3(f_2(f_1(z))) = \frac{1-z}{3+z}$ is also a linear fractional transformation. Moreover, f maps the disk $B_2(-1)$ onto the right half-plane. This composition is demonstrated in Figure ??.

Figure 2.2: To map a disk to the half-plane, it is always advantageous to map the given disk to the unit disk and then use the transformation ϕ in Example ??.

Before we explore more examples of similar compositions, we introduce propositions that are useful for explicitly finding linear fractional transformations given information on three distinct points and their images.

Lemma 2.9 (Fixed points). *A linear fractional transformation ϕ can have at most two fixed points in the complex plane, unless $\phi(z) = z$, in which case all points are fixed points.*

Proof. Let ϕ be a linear fractional transformation. We consider all solutions to the equation $z = \frac{az + b}{cz + d}$. Two cases arise. First, let's assume $c = 0$. Then, we will solve $z = \frac{az + b}{d}$. Here $z = \frac{b}{d - a}$. Next, let's assume $c \neq 0$. Then, $z = \frac{az + b}{cz + d}$ implies that $z(cz + d) = az + b$. This further implies that $cz^2 + (d - a)z - b = 0$, which has at most two solutions. In either case, we see that z has at most two solutions. Therefore, there are at most two fixed points for ϕ unless ϕ is the identity map, in which case all points are fixed points. ■

Lemma 2.10 (Uniqueness of a linear fractional transformation). *Let z_1, z_2 , and z_3 be three distinct points, and let w_1, w_2 , and w_3 be three distinct points (we allow ∞). If there is a linear fractional transformation mapping z_j to w_j for $j = 1, 2, 3$, then it is unique.*

Proof. Let f and g be two linear fractional transformations mapping z_j to w_j for $j = 1, 2, 3$. Notice that g^{-1} exists and is also a linear fractional transformation mapping w_j to z_j for $j = 1, 2, 3$ by Proposition 2.1. Notice then that the composition $g^{-1} \circ f$ is also a linear fractional transformation by Proposition 2.3. Then, we have

$$g^{-1}(f(z_j)) = g^{-1}(w_j) = z_j.$$

Since this is true for all $j = 1, 2, 3$, then we conclude that $g^{-1} \circ f$ has three fixed points. By Lemma ??, we conclude that $g^{-1}(f(z)) = z$ and so $f = g$. ■

Proposition 2.11. *There is a unique linear fractional transformation w that maps three distinct points z_1, z_2 , and z_3 onto three distinct points w_1, w_2 , and w_3 . The mapping w is given implicitly (as a function of z) by*

$$\frac{z - z_1}{z - z_3} \frac{z_2 - z_3}{z_2 - z_1} = \frac{w - w_1}{w - w_3} \frac{w_2 - w_3}{w_2 - w_1}. \quad (2.3)$$

Proof. We can see that w given above is indeed a linear fractional transformation if we solve for w . To do so, let $\alpha = \frac{z_2 - z_3}{z_2 - z_1}$ and $\beta = \frac{w_2 - w_3}{w_2 - w_1}$. Note $\alpha, \beta \neq 0$ as z_j and w_j are unique for $j = 1, 2, 3$. Solving for w , we obtain

$$w = \frac{(\alpha w_3 - \beta w_1)z + (\beta w_1 z_3 - \alpha z_1 w_3)}{(\alpha - \beta)z + (\beta z_3 - \alpha z_1)}.$$

Let $a = \alpha w_3 - \beta w_1$, $b = \beta w_1 z_3 - \alpha z_1 w_3$, $c = \alpha - \beta$, and $d = \beta z_3 - \alpha z_1$. We have that $a, b, c, d \in \mathbb{C}$ and in fact

$$ad - bc = \alpha\beta(w_1 - w_3)(z_1 - z_3) \neq 0$$

by the uniqueness of the points. Using our explicit formula for w , we see $w(z_1) = w_1$, $w(z_2) = w_2$, and $w(z_3) = w_3$. Uniqueness follows by Lemma ??.

Proposition 2.12. *Let z_1, z_2 , and z_3 be three distinct points. There is a unique linear fractional transformation w that maps z_1 and z_2 onto two distinct points w_1 and w_2 , and maps z_3 to ∞ . The mapping w is given implicitly (as a function of z) by*

$$\frac{z - z_1}{z - z_3} \frac{z_2 - z_3}{z_2 - z_1} = \frac{w - w_1}{w_2 - w_1}.$$

Proof. We begin by solving

$$\frac{z - z_1}{z - z_3} \frac{z_2 - z_3}{z_2 - z_1} = \frac{w - w_1}{w_2 - w_1}$$

for w . It follows that

$$\begin{aligned} w &= \frac{(z - z_1)(z_2 - z_3)(w_2 - w_1)}{(z - z_3)(z_2 - z_1)} + w_1 \\ &= \frac{z[(z_2 - z_3)(w_2 - w_1) + w_1(z_2 - z_1)] - z_1(z_2 - z_3)(w_2 - w_1) - w_1 z_3(z_2 - z_1)}{z(z_2 - z_1) - z_3(z_2 - z_1)}. \end{aligned}$$

Notice that w is in the form $w = \frac{az + b}{cz + d}$. Here we have

$$a = (z_2 - z_3)(w_2 - w_1) + w_1(z_2 - z_1)$$

$$b = -z_1(z_2 - z_3)(w_2 - w_1) - w_1 z_3(z_2 - z_1)$$

$$c = z_2 - z_1$$

$$d = -z_3(z_2 - z_1).$$

Notice that $a, b, c, d \in \mathbb{C}$, so we just need to verify that $ad - bc \neq 0$.

$$ad - bc = (z_1 - z_3)(z_2 - z_1)(z_2 - z_3)(w_2 - w_1) \neq 0$$

by the uniqueness of the points. Next, we verify that w maps z_1 and z_2 onto two distinct points w_1 and w_2 , and maps z_3 to ∞ . When $z = z_1$ and $w = w_1$, the equation $\frac{z - z_1}{z - z_3} \frac{z_2 - z_3}{z_2 - z_1} = \frac{w - w_1}{w_2 - w_1}$ results in $0 = 0$. When $z = z_2$ and $w = w_2$, the equation

$\frac{z - z_1}{z - z_3} \frac{z_2 - z_3}{z_2 - z_1} = \frac{w - w_1}{w_2 - w_1}$ results in $1 = 1$. The fact that

$$\lim_{z \rightarrow z_3} \frac{z - z_1}{z - z_3} \frac{z_2 - z_3}{z_2 - z_1} = \infty \text{ and } \lim_{w \rightarrow \infty} \frac{w - w_1}{w_2 - w_1} = \infty$$

tells us that z_3 maps to ∞ . Uniqueness follows by Lemma ??.

■

Proposition 2.13. *Let z_1 and z_2 be two distinct points, and w_1, w_2 , and w_3 be three distinct points. There is a unique linear fractional transformation w that maps z_1 to w_1 , z_2 to w_2 , and ∞ to w_3 . The mapping w is implicitly given (as a function of z) by*

$$\frac{z - z_1}{z_2 - z_1} = \frac{w - w_1}{w - w_3} \frac{w_2 - w_3}{w_2 - w_1}.$$

Proof. We begin by solving

$$\frac{z - z_1}{z_2 - z_1} = \frac{w - w_1}{w - w_3} \frac{w_2 - w_3}{w_2 - w_1}$$

for w . Solving for w yields the following equation

$$w = \frac{-w_3 z(w_2 - w_1) + w_3 z_1(w_2 - w_1) + w_1(z_2 - z_1)(w_2 - w_3)}{-z(w_2 - w_1) + z_1(w_2 - w_1) + (z_2 - z_1)(w_2 - w_3)}.$$

Notice that w is in the form $w = \frac{az+b}{cz+d}$. Here we have

$$a = -w_3(w_2 - w_1)$$

$$b = w_3 z_1(w_2 - w_1) + w_1(z_2 - z_1)(w_2 - w_3)$$

$$c = -(w_2 - w_1)$$

$$d = z_1(w_2 - w_1) + (z_2 - z_1)(w_2 - w_3).$$

Notice $a, b, c, d \in \mathbb{C}$, so we just need to verify that $ad - bc \neq 0$. Observe that

$$ad - bc = (w_3 - w_1)(w_2 - w_1)(w_2 - w_3)(z_2 - z_1) \neq 0$$

by the uniqueness of the points. Next, we will verify that w maps z_1 to w_1 , z_2 to w_2 ,

and ∞ to w_3 . When $z = z_1$ and $w = w_1$, then the equation $\frac{z-z_1}{z_2-z_1} = \frac{w-w_1}{w-w_3} \frac{w_2-w_3}{w_2-w_1}$ yields

$0 = 0$. When $z = z_2$ and $w = w_2$, then the equation $\frac{z-z_1}{z_2-z_1} = \frac{w-w_1}{w-w_3} \frac{w_2-w_3}{w_2-w_1}$ yields $1 = 1$.

The fact that

$$\lim_{z \rightarrow \infty} \frac{z - z_1}{z_2 - z_1} = \infty \text{ and } \lim_{w \rightarrow w_3} \frac{w - w_1}{w - w_3} \frac{w_2 - w_3}{w_2 - w_1} = \infty$$

tells us that ∞ maps to w_3 . Uniqueness follows by Lemma ??.

■

We will now consider applications of these results in the following examples.

Example 2.14. We supply the formulas of the analytic mappings in each sequence of mappings shown in Figure ?? and derive the given formula for the final composite mapping.

Figure 2.3: Sequence of mappings for Example ??.

Solution. First, we want to shift our graph down by $1 + i$ units so that the circle is centered at the origin. This mapping is given by $f_1(z) = z - 1 - i$. Next, we use the inversion map $f_2(w_1) = \frac{1}{w_1}$ to map to the interior of the unit circle. Lastly, $f_3(w_2) = i\frac{1-w_2}{1+w_2}$ by Example ??. Furthermore, the composition of these maps is given by

$$g(z) = f_3(f_2(f_1(z))) = i\frac{z-2-i}{z-i}.$$

Example 2.15. We supply the formulas of the analytic mappings in each sequence of mappings shown in Figure ?? and derive the given formula for the final composite mapping.

Figure 2.4: Sequence of mappings for Example ??.

Solution. To construct f_1 , we use that fact that $-2 \mapsto 0$, $0 \mapsto 1$, and $2 \mapsto \infty$. Using Proposition ??, we get that $f_1(z) = \frac{-z-2}{z-2}$. Next, we have that $f_2(w_1) = w_1^6$ maps the region in the second picture to the upper half-plane by DeMoivre's identity. Lastly, $f_3(w_2) = \frac{i-w_2}{i+w_2}$ by Example ??. Furthermore, the composition of these maps is given by

$$g(z) = f_3(f_2(f_1(z))) = \frac{i(z-2)^6 - (z+2)^6}{i(z-2)^6 + (z+2)^6}.$$

Example 2.16. We supply the formulas of the analytic mappings in each sequence of mappings shown in Figure ?? and derive the given formula for the final composite mapping.

Figure 2.5: Sequence of mappings for Example ??.

Solution. To construct f_1 , notice that $2i \mapsto -\frac{i}{2}$, $-2 \mapsto -\frac{1}{2}$, and $0 \mapsto \infty$. Using Proposition ??, we get that $f_1(z) = \frac{1}{z}$. Next, we have $f_2(w_1) = w_1 + \frac{1+i}{2}$ as a shift of $\frac{1+i}{2}$ was preformed to transition between the second and third regions. Lastly, $f_3(w_2) = (-iw_2)^{\frac{2}{3}}$ by DeMoivre's identity. Furthermore, the composition of these maps is given by

$$g(z) = f_3(f_2(f_1(z))) = \left(\frac{-i}{z} + \frac{1-i}{2} \right)^{\frac{2}{3}}.$$

Example 2.17. We supply the formulas of the analytic mappings in each sequence of mappings shown in Figure ?? and derive the given formula for the final composite mapping.

Solution. To construct f_1 , notice that $-2 \mapsto \frac{3i}{2}$, $-1 \mapsto i$, and $1 \mapsto 0$. Using Proposition ??, we obtain $f_1(z) = i\frac{1-z}{2}$. To construct f_2 , notice that $i \mapsto 0$, $0 \mapsto 1$, and $\infty \mapsto -1$. Using Proposition ??, we obtain $f_2(w_1) = -\frac{w_1-i}{w_1+i}$. Thus, the composition of these functions is given by $g(z) = f_2(f_1(z)) = \frac{z+1}{-z+3}$.

Example 2.18. We supply the formulas of the analytic mappings in each sequence of mappings shown in Figure ?? and derive the given formula for the final composite mapping.

Figure 2.7: Sequence of mappings for Example ??.

Solution. First, $f_1(z) = \frac{1}{4}z^2$ maps the quarter circle with radius two to the upper half-circle with radius one as a result of DeMoivre's identity and a shrink. To construct f_2 , notice that $1 \mapsto 0$, $0 \mapsto i$, and $-1 \mapsto \infty$. By Proposition ??, we have $f_2(w_1) = -i\frac{w_1-1}{w_1+1}$. The composition of these maps is given by

$$g(z) = f_2(f_1(z)) = i \left(\frac{-z^2 + 4}{z^2 + 4} \right).$$

Example 2.19. We supply the formulas of the analytic mappings in each sequence of mappings shown in Figure ?? and derive the given formula for the final composite mapping.

Figure 2.8: Sequence of mappings for Example ??.

Solution. We have $f_1(z) = e^z$ and $f_2(w_1) = iw_1$. The composition of these maps is given by $g(z) = f_2(f_1(z)) = ie^z$.

These are just a few examples of how one can map a specific region to another region, which may be more useful for solving problems on, through a sequence of conformal maps. The next interesting situation we may encounter is a region bounded by two non-intersecting circles, C_2 and C_1 , such that C_1 is in the interior of C_2 . We would prefer to work with circles that have the same center, so the challenge becomes mapping the region described above to an annulus.

Proposition 2.20. *Let C_2 be the unit circle and let C_1 be a circle on the interior of C_2 centered on the x -axis with x -intercepts $(a, 0)$ and $(b, 0)$ with $|a| < b < 1$. Let Ω be the region bounded by C_1 and C_2 . Then, $\phi(z) = \frac{z-\alpha}{1-\alpha z}$ with $\alpha = \frac{1+ab}{a+b} - \sqrt{\left(\frac{1+ab}{a+b}\right)^2 - 1}$ will map C_2 onto C_2 , C_1 onto a circle centered at the origin with radius $r = \phi(b)$,*

and the region between C_2 and C_1 onto the annular region bounded $\phi[C_1]$ and the unit circle.

Proof. We will show that we can choose an α such that $-1 < \alpha < 1$ and $\phi_\alpha(z) = \frac{z-\alpha}{1-\bar{\alpha}z}$ maps the circle C_1 onto a circle centered at the origin. It is known that this map maps the unit circle onto itself, so C_2 is mapped onto itself under ϕ_α [?]. It suffices to show choose an α so that $\phi_\alpha(a) = -\phi_\alpha(b)$. The reason being is that we would like for $\phi_\alpha(a)$ and $\phi_\alpha(b)$ to serve as endpoints for the diameter of our new circle $\phi[C_1]$ centered at the origin. This would imply that $\phi_\alpha(a) = -\phi_\alpha(b)$. With this condition, we have

$$\frac{a-\alpha}{1-\alpha a} = -\frac{b-\alpha}{1-\alpha b}.$$

Moreover, we have

$$\alpha^2 - 2\frac{1+ab}{a+b}\alpha + 1 = 0.$$

It's important to note that $a+b \neq 0$ from the assumption that $|a| < b < 1$. Using the quadratic equation to solve for α , we obtain

$$\alpha_1 = \frac{1+ab}{a+b} + \sqrt{\left(\frac{1+ab}{a+b}\right)^2 - 1}, \alpha_2 = \frac{1+ab}{a+b} - \sqrt{\left(\frac{1+ab}{a+b}\right)^2 - 1}.$$

We next proceed to show that $\alpha_1 > 1$ and $0 < \alpha_2 < 1$, which means we must choose α_2 in constructing ϕ_α . To do so, we will first show that $1+ab \geq a+b$ given the assumptions that $|a| < 1$, $0 < b < 1$ and $0 \leq |a| < b < 1$. Since $1-b > 0$, if $0 \leq a < 1$, then multiplying $1-b$ by a , we get $1-b > a(1-b)$. If $-1 < a < 0$, then $1-b > a(1-b)$, because the right side is negative, while the left side is positive. In all cases, $1+ab > a+b > 0$. Consequently, $\left(\frac{1+ab}{a+b}\right)^2 > 1$; hence the discriminant of the quadratic equation in α in part is positive and so we have two distinct roots, α_1 and

α_2 . From the inequality $1 + ab > a + b > 0$, we conclude that $\frac{1+ab}{a+b} > 1$ and so $\alpha_1 > 1$. The product of $\alpha_1 \cdot \alpha_2 = 1$, as can be checked directly or by using a well-known property that the product of the roots of the quadratic equation $ax^2 + bx + c = 0$ is $\frac{c}{a}$. Since $1 < \alpha_1$, we conclude that $0 < \alpha_2 < 1$, in order to have the equality $\alpha_1 \cdot \alpha_2 = 1$. We conclude that $\phi(z) = \frac{z-\alpha}{1-\alpha z}$ with $\alpha = \frac{1+ab}{a+b} - \sqrt{\left(\frac{1+ab}{a+b}\right)^2 - 1}$ will map C_2 onto C_2 , C_1 onto a circle centered at the origin with radius $r = \phi(b)$, and the region between C_2 and C_1 onto the annular region bounded by $\phi[C_1]$ and the unit circle. ■

Note: If C_2 is not the unit circle, we can find a mapping that shifts and scales C_2 by the appropriate increments so that we obtain the unit circle. After doing so, we can apply the process established in Proposition ??.

Example 2.21. We derive the linear fractional transformation that maps the shaded region between the two given circles onto an annular region centered at the origin.

Figure 2.9: Here $a = -\frac{1}{4}$, $b = \frac{4}{7}$.

Solution. Note that $a = -\frac{1}{4}$ and $b = \frac{4}{7}$ satisfies $|a| < b < 1$, so we use the linear fractional transformation from Proposition ??. Using the formula for α , we obtain $\alpha = \frac{8-\sqrt{55}}{3}$. Thus, $\phi(z) = \frac{3z-(8-\sqrt{55})}{3-(8-\sqrt{55})z}$ from Proposition ??.

Example 2.22. We supply the formulas of the analytic mappings in each sequence of mappings shown in Figure ?? and derive the given formula for the final composite mapping $w = f(z)$.

Figure 2.10: Sequence of mappings for Example ??.

Solution. The first map from the z -plane to the w_1 -plane is given by $f_1(z) = \frac{1}{z}$. The second map from the w_1 -plane to the w_2 -plane is simply a counterclockwise rotation

by $\frac{\pi}{4}$ and is given by $f_2(w_1) = (\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2})w_1$. The last map from w_2 -plane to the w -plane is of the type described in Proposition ?? . Here $\alpha = \frac{1}{2}$ and so $f_3(w_2) = \frac{2w_2-1}{2-w_2}$. The composition $g(z) = f_3(f_2(f_1(z))) = \frac{2(\sqrt{2}+i\sqrt{2})-2z}{4z-(\sqrt{2}+i\sqrt{2})}$.

Chapter 3

The Schwarz-Christoffel Transformation

In this chapter, we describe another particular type of one-to-one, conformal mapping, the Schwarz-Christoffel transformation. The Schwarz-Christoffel transformation will allow us to map the upper half-plane onto a polygonal region.

Suppose Ω is a region in the w -plane, bounded by a positively oriented polygonal path P with n sides. We will let $w_1, w_2, \dots, w_n$ denote the vertices of P counted as the path P is traced with positive orientation. Figure ?? illustrates the case when $n = 5$. If P is bounded, then the point w_n is taken to be the initial and terminal point of the closed path P . If P is unbounded, we specify $w_n = \infty$ and view P as a polygon with $n - 1$ vertices $w_1, w_2, \dots, w_{n-1}$. This is illustrated in Figure ?. Additionally, we let θ_j denote the angle created as we turn the corner of the polygon at w_j . We choose $0 < |\theta_j| < \pi$ for $j = 1, \dots, n$. A positive value corresponds to a left turn; meanwhile, a negative value corresponds to a right turn. We see in Figure ?? that θ_2 is the only negative angle.

Figure 3.1: Positively oriented polygonal boundary with corner angles measured from the outside.

Figure 3.2: Unbounded polygonal region with 4 sides and 3 vertices.

Theorem 3.1 (Schwarz-Christoffel Transformation). *Let Ω be a region bounded by*

a polygonal path P with vertices at w_j (counted consecutively) and corresponding exterior angles θ_j . Then there is a one-to-one conformal mapping f of the upper half-plane onto Ω , such that

$$f'(z) = A(z - x_1)^{-\frac{\theta_1}{\pi}}(z - x_2)^{-\frac{\theta_2}{\pi}} \cdots (z - x_{n-1})^{-\frac{\theta_{n-1}}{\pi}}, \quad (3.1)$$

where A is a constant, the x_j 's are real and satisfy $x_1 < x_2 < \cdots < x_{n-1}$, $f(x_j) = w_j$, $\lim_{z \rightarrow \infty} f(z) = w_n$, and the complex powers are defined by their principal branches.

The points x_j on the x -axis are the pre-images of the vertices of the polygonal path P in the w -plane. We may choose two of the x_j 's arbitrarily as long as they are arranged in ascending order. Similar to our discussion in the previous chapter we denote $\lim_{z \rightarrow \infty} f(z) = w_n$ as $f(\infty) = w_n$. If P is unbounded, we have $f(\infty) = \infty$.

Definition 3.2. The mapping f whose derivative is given in Equation ?? is called a *Schwarz-Christoffel transformation*, which is named after the German mathematicians Karl Herman Amandus Schwarz (1843-1921) and Elwin Bruno Christoffel (1829-1900). Since f is an antiderivative of Equation ??, we can write

$$f(z) = A \int (z - x_1)^{-\frac{\theta_1}{\pi}} (z - x_2)^{-\frac{\theta_2}{\pi}} \cdots (z - x_{n-1})^{-\frac{\theta_{n-1}}{\pi}} dz + B. \quad (3.2)$$

Here the constants A and B depend on the polygonal path P .

Proving Theorem ?? is quite difficult, so we provide a sketch of the proof and a proof in the case of the upper half-plane being mapped onto an arbitrary triangle. The sketch of the general proof for a closed polygon requires the following lemma, which we prove first.

Lemma 3.3 (Closure of the polygon). *Let P be a closed polygon with $\theta_1 + \cdots + \theta_n = 2\pi$ and f be given as in Equation ?. Then, $f(z) \rightarrow w_n$ as $z \rightarrow \infty$.*

Proof. Suppose $\theta_1 + \theta_2 + \dots + \theta_n = 2\pi$. We can safely assume $\theta_n < \pi$ by the assumption that we can choose $0 < |\theta_j| < \pi$ for $j = 1, \dots, n$. It follows that

$$2\pi = \theta_1 + \theta_2 + \dots + \theta_n < \theta_1 + \theta_2 + \dots + \theta_{n-1} + \pi,$$

so we conclude that $\pi < \theta_1 + \theta_2 + \dots + \theta_{n-1}$ by subtracting π from both sides of the inequality above. We conclude that $\theta_1 + \theta_2 + \dots + \theta_{n-1} > \pi$. We let $\beta_j = \frac{\theta_j}{\pi}$.

Because A and B in Equation ?? just dilate, rotate, and translate the mapping, A and B do not affect the convergence of f or the closure of the polygon. Hence, we can choose $A = 1$ and $B = 0$. Let $x_0 = 1 + \max_{1 \leq j \leq n-1} x_j$ and set

$$f(z) = \int_{[x_0, z]} \frac{d\zeta}{(\zeta - x_1)^{\beta_1} (\zeta - x_2)^{\beta_2} \dots (\zeta - x_{n-1})^{\beta_{n-1}}}.$$

Next, let $x_0 = 1 + \max_{1 \leq j \leq n-1} x_j$ and $z = x$ be real. We will use the limit comparison test for the integrand to show that

$$\lim_{y \rightarrow \infty} f(y) = \int_{x_0}^{\infty} \frac{dx}{(x - x_1)^{\beta_1} (x - x_2)^{\beta_2} \dots (x - x_{n-1})^{\beta_{n-1}}}$$

is finite. Let $a = \max_{1 \leq j \leq n-1} x_j$ and $b = \min_{1 \leq j \leq n-1} x_j$. Observe that

$$\begin{aligned} x - a + a - b &= (x - a) \left(1 + \frac{a - b}{x - a} \right) \\ &\leq (x - a)(1 + a - b) \end{aligned}$$

because $x - a \geq 1$. This observation gives us

$$(1 + a - b)(x - a) \geq x - a + a - b = x - b \geq x - x_j \geq x - a$$

for all $1 \leq j \leq n - 1$. We see that $(x - x_j)^{\beta_j} \approx (x - a)^{\beta_j}$ for all β_j . Using this, we have

$$\int_{x_0}^{\infty} \frac{dx}{(x - x_1)^{\beta_1} (x - x_2)^{\beta_2} \dots (x - x_{n-1})^{\beta_{n-1}}} \leq C \int_{x_0}^{\infty} \frac{dx}{(x - a)^{\beta_1 + \beta_2 + \dots + \beta_{n-1}}},$$

where C is a constant. Specifically, we have

$$C = \prod_{j:\beta_j>0} (1+a-b)^{\beta_j}.$$

Notice that

$$\beta_1 + \beta_2 + \cdots + \beta_{n-1} = \frac{\theta_1 + \theta_2 + \cdots + \theta_{n-1}}{\pi} > \frac{\pi}{\pi} = 1$$

from our previous work above. Therefore, after a change in variables, we see that the integral on the right converges.

Lastly, let

$$w_n = \lim_{y \rightarrow \infty} f(y) = \int_{x_0}^{\infty} \frac{dx}{(x-x_1)^{\beta_1}(x-x_2)^{\beta_2} \cdots (x-x_{n-1})^{\beta_{n-1}}}$$

from above. We want to show that $\lim_{|z| \rightarrow \infty} f(z) = w_n$ for any $z \in \mathbb{C}$. We will prove this for $z = Re^{i\theta}$ where $R > 0$ and $0 \leq \theta \leq \pi$. A similar argument will hold for $z = Re^{i\theta}$ where $R > 0$ and $\pi \leq \theta \leq 2\pi$. First, observe that

$$\begin{aligned} |f(Re^{i\theta}) - f(R)| &= \left| \int_{[x_0, Re^{i\theta}]} \frac{dx}{(x-x_1)^{\beta_1}(x-x_2)^{\beta_2} \cdots (x-x_{n-1})^{\beta_{n-1}}} \right. \\ &\quad \left. - \int_{[x_0, R]} \frac{dx}{(x-x_1)^{\beta_1}(x-x_2)^{\beta_2} \cdots (x-x_{n-1})^{\beta_{n-1}}} \right| \\ &= \left| \int_{[x_0, Re^{i\theta}]} \frac{dx}{(x-x_1)^{\beta_1}(x-x_2)^{\beta_2} \cdots (x-x_{n-1})^{\beta_{n-1}}} \right. \\ &\quad \left. + \int_{[R, x_0]} \frac{dx}{(x-x_1)^{\beta_1}(x-x_2)^{\beta_2} \cdots (x-x_{n-1})^{\beta_{n-1}}} \right| \\ &= \left| \int_{[R, Re^{i\theta}]} \frac{dx}{(x-x_1)^{\beta_1}(x-x_2)^{\beta_2} \cdots (x-x_{n-1})^{\beta_{n-1}}} \right| \\ &\leq R\pi M(R), \end{aligned}$$

where $M(R)$ is the maximum of the absolute value of the integrand on the upper semicircle of radius R . Also, the length of the segment joining R to $Re^{i\theta}$ is largest when it is the circumference of the upper semicircle or $R\pi$. Next, we claim that

$\lim_{R \rightarrow \infty} M(R)R^{\beta_1 + \dots + \beta_{n-1}} = 1$. Notice that for $R > \max_{1 \leq j \leq n-1} |x_j|$ we have

$$\begin{aligned}
M(R) &= \sup_{|z|=R} \left| \frac{1}{(z - x_1)^{\beta_1} \dots (z - x_{n-1})^{\beta_{n-1}}} \right| \\
&= \sup_{|z|=R} \left[\prod_{\substack{1 \leq j \leq n-1 \\ \beta_j \geq 0}} \frac{1}{|z - x_j|^{\beta_j}} \prod_{\substack{1 \leq j \leq n-1 \\ \beta_j < 0}} \frac{1}{|z - x_j|^{\beta_j}} \right] \\
&= \sup_{|z|=R} \left[\prod_{\substack{1 \leq j \leq n-1 \\ \beta_j \geq 0}} \frac{1}{|z - x_j|^{\beta_j}} \prod_{\substack{1 \leq j \leq n-1 \\ \beta_j < 0}} |z - x_j|^{\beta_j} \right] \\
&= \prod_{\substack{1 \leq j \leq n-1 \\ \beta_j \geq 0}} \frac{1}{(R - x_j)^{\beta_j}} \prod_{\substack{1 \leq j \leq n-1 \\ \beta_j < 0}} (R + x_j)^{|\beta_j|},
\end{aligned}$$

since the first product becomes maximum for $z = R$ and the second product becomes maximum for $z = -R$ and both of these numbers lie on the circle $|z| = R$. This implies that

$$\lim_{R \rightarrow \infty} M(R)R^{\beta_1 + \dots + \beta_{n-1}} = \lim_{R \rightarrow \infty} \prod_{\substack{1 \leq j \leq n-1 \\ \beta_j \geq 0}} \frac{R^{|\beta_j|}}{(R - x_j)^{\beta_j}} \prod_{\substack{1 \leq j \leq n-1 \\ \beta_j < 0}} \frac{(R + x_j)^{|\beta_j|}}{R^{|\beta_j|}}.$$

But, we have $\beta_i \geq 0$

$$\lim_{R \rightarrow \infty} \frac{R^{\beta_i}}{(R - x_i)^{\beta_i}} = 1$$

and for $\beta_j < 0$,

$$\lim_{R \rightarrow \infty} \frac{(x_j + R)^{|\beta_j|}}{R^{|\beta_j|}} = 1.$$

We conclude that $\lim_{R \rightarrow \infty} M(R) = 1$. Note that

$$RM(R) = R^{1 - (\beta_1 + \dots + \beta_{n-1})} (M(R)R^{\beta_1 + \dots + \beta_{n-1}}).$$

The fact that $\beta_1 + \dots + \beta_{n-1} > 1$ by our discussion in part (c) implies that $RM(R) \rightarrow 0$ as $R \rightarrow \infty$. This implies that $|f(Re^{i\theta}) - f(R)| \rightarrow 0$ uniformly in θ as $R \rightarrow \infty$. The fact that $f(R) \rightarrow w_n$ as $R \rightarrow \infty$ implies that $f(z) \rightarrow w_n$ as $|z| \rightarrow \infty$. ■

Proof. Let f be such a function whose derivative is given by Equation ???. Let $w_j = f(x_j)$ for $1 < j < n - 1$ with $x_1 < x_2 < \cdots < x_{n-1}$. For any point $x_0 \in \mathbb{R}$, the slope of the line segment containing $f(x_0)$ is given by $\arg(f'(x_0))$ by Theorem ??. By the properties of the argument, we have

$$\arg(f'(z)) = \arg(A) - \frac{\theta_1}{\pi} \arg(z - x_1) - \frac{\theta_2}{\pi} \arg(z - x_2) - \cdots - \frac{\theta_{n-1}}{\pi} \arg(z - x_{n-1}).$$

For $z \in \mathbb{R}$ with $z < x_1$, we will have $z < x_j$ for all $1 < j < n - 1$. This implies that $z - x_j < 0$ for all $1 < j < n - 1$, which means $\arg(z - x_j) = \pi$ for all $1 < j < n - 1$. Thus, $\arg(f'(z)) = \arg(A) - (\theta_1 + \theta_2 + \cdots + \theta_{n-1})$. If $w_n = f(\infty)$, all the points in the interval $(-\infty, x_1)$ are mapped onto a line segment starting with w_n and ending with $w_1 = f(x_1)$ and at an angle given by $\arg(f'(z)) = \arg(A) - (\theta_1 + \theta_2 + \cdots + \theta_{n-1})$. For the points in the interval (x_1, x_2) , we have $z - x_1 > 0$ and $z - x_j < 0$ for all $2 < j < n - 1$. Then, $\arg(z - x_1) = 0$ and $\arg(z - x_j) = \pi$ for $2 < j < n - 1$. Here we have $\arg(f'(z)) = \arg(A) - (\theta_2 + \cdots + \theta_{n-1})$. At the point x_1 , the path turns by an angle of θ_1 and all the points in (x_1, x_2) are mapped onto the line segment with initial point w_1 and terminal point $w_2 = f(x_2)$. We continue this process until we have the points in the interval (x_{n-1}, ∞) being mapped onto a line segment with initial point w_{n-1} and at angle $\arg(f'(z)) = \arg(A)$. If we have bounded polygon, this line segment will connect back to w_n by Lemma ??. The resulting polygon is closed and $\theta_1 + \cdots + \theta_n = 2\pi$. This shows that f takes the real line onto a polygon with vertices $w_j = f(x_j)$ and exterior angles θ_j . Since the upper half-plane is on our left as we traverse the real line, we know that interior our the resulting polygon P will be the left as we traverse the polygon with positive orientation. ■

Proof. We prove Theorem ?? in the case where the upper half-plane is mapped to

an arbitrary triangle. The argument we provide is based on geometric considerations and can be made to work for general polygons.

Figure 3.3: Here Ω is the upper half-plane.

Figure 3.4: The image of Ω under a Schwarz-Christoffel transformation f is the enclosed triangle.

Notice that $\arg(f'(y_2)), \arg(f'(y_3)) > 0$; meanwhile, $\arg(f'(y_1)) < 0$. Let

$$\varphi_1 = \arg(f'(y_1)) + \pi$$

$$\varphi_2 = \arg(f'(y_2))$$

$$\varphi_3 = \arg(f'(y_3))$$

to ensure that $\varphi_j > 0$ for all $j = 1, 2, 3$. We want to verify the following

$$(1) \quad \arg(f'(y_1)) = \arg(A) - (\theta_1 + \theta_2)$$

$$(2) \quad \arg(f'(y_2)) = \arg(A) - \theta_2$$

with the assumption that $\arg(f'(y_3)) = \arg(A)$. For (2), notice that $\varphi_3 = \varphi_2 + \theta_2$ by the fact that the an exterior angle is the sum of the two interior angles. Thus, we have established (2). For (1), we will first find ω . Here

$$\omega = \pi - (\pi - \theta_1 + \pi - \theta_2) = \theta_1 + \theta_2 - \pi.$$

This implies $\varphi_3 = \varphi_1 + (\theta_1 + \theta_2 - \pi)$, which establishes (1). ■

Example 3.4. We find the Schwarz-Christoffel transformation of the upper half-plane onto the region given by Figure ?? and Figure ??.

Figure 3.5: Part (a).

Figure 3.6: Part (b).

Solution

(a) We have that $x_1 = -1$, $x_2 = 1$, $\theta_1 = -\pi$, and $\theta_2 = \frac{\pi}{2}$. Using Equation ??, we obtain

$$\begin{aligned} f(z) &= A \int (z+1)(z-1)^{-\frac{1}{2}} dz + B \\ &= A \int \frac{z+1}{(z-1)^{\frac{1}{2}}} dz + B \\ &= A \int (z-1)^{\frac{1}{2}} + \frac{2}{(z-1)^{\frac{1}{2}}} dz + B \\ &= A \left[\frac{2}{3}(z-1)^{\frac{3}{2}} + 4(z-1)^{\frac{1}{2}} \right] + B. \end{aligned}$$

The fact that $f(1) = 1$ implies $B = 1$. The fact that $f(-1) = 0$ implies that $0 = A \left(\frac{2^{\frac{7}{2}}i}{3} \right) + 1$. Thus, $A = \frac{3i}{2^{\frac{7}{2}}}$. We conclude that

$$f(z) = \frac{3i}{2^{\frac{7}{2}}} \left[\frac{2}{3}(z-1)^{\frac{3}{2}} + 4(z-1)^{\frac{1}{2}} \right] + 1.$$

(b) We have that $x_1 = -1$, $x_2 = 1$, $\theta_1 = -\frac{\pi}{2}$, and $\theta_2 = \frac{\pi}{2}$. Using Equation ??, we have

$$\begin{aligned} f(z) &= A \int (z+1)^{\frac{1}{2}}(z-1)^{-\frac{1}{2}} dz + B \\ &= A \int \frac{(z+1)^{\frac{1}{2}}}{(z-1)^{\frac{1}{2}}} dz + B \\ &= A \int \frac{z}{i(1-z^2)^{\frac{1}{2}}} + \frac{1}{i(1-z^2)^{\frac{1}{2}}} dz + B \end{aligned}$$

$$\begin{aligned}
&= A[i(1 - z^2)^{\frac{1}{2}} - i \sin^{-1}(z)] + B \\
&= Ai[(1 - z^2)^{\frac{1}{2}} - \sin^{-1}(z)] + B.
\end{aligned}$$

The fact that $f(-1) = 0$ implies that $\frac{A\pi i}{2} + B = 0$. The fact that $f(1) = -i$ implies that $-\frac{A\pi i}{2} + B = -i$. Solving this system of equations yields $B = -\frac{i}{2}$ and $A = \frac{1}{\pi}$. Thus, we have

$$f(z) = \frac{i}{\pi}[(1 - z^2)^{\frac{1}{2}} - \sin^{-1}(z)] - \frac{i}{2}.$$

Example 3.5. We find the Schwarz-Christoffel transformation of the upper half-plane onto the region given by Figure ?? and Figure ??.

Figure 3.7: Part (a).

Solution

(a) We have that $x_1 = -1$, $x_2 = 1$, $\theta_1 = \frac{\pi}{2}$, and $\theta_2 = -\frac{\pi}{2}$. Using Equation ??, we have

$$\begin{aligned}
f(z) &= A \int (z+1)^{-\frac{1}{2}}(z-1)^{\frac{1}{2}} dz + B \\
&= A \int \frac{(z-1)^{\frac{1}{2}}}{(z+1)^{\frac{1}{2}}} dz + B \\
&= A \int \frac{z}{i(1-z^2)^{\frac{1}{2}}} - \frac{1}{i(1-z^2)^{\frac{1}{2}}} dz + B \\
&= A[i(1-z^2)^{\frac{1}{2}} + i \sin^{-1}(z)] + B \\
&= Ai[(1-z^2)^{\frac{1}{2}} + \sin^{-1}(z)] + B.
\end{aligned}$$

The fact that $f(-1) = 0$ implies that $-\frac{A\pi i}{2} + B = 0$. The fact that $f(1) = i$ implies that $\frac{A\pi i}{2} + B = i$. Solving the system of equations yields $B = \frac{i}{2}$ and

$A = \frac{1}{\pi}$. Thus, we have that

$$f(z) = \frac{i}{\pi} \left[(1 - z^2)^{\frac{1}{2}} + \sin^{-1}(z) \right] + \frac{i}{2}.$$

(b) We have that $x_1 = -1$, $x_2 = 1$, $\theta_1 = \pi$, and $\theta_2 = -\frac{\pi}{2}$. Using Equation ??, we have

$$\begin{aligned} f(z) &= A \int (z+1)^{-1} (z-1)^{\frac{1}{2}} dz + B \\ &= A \int \frac{1}{\sqrt{z-1}} - \frac{2}{(\sqrt{z-1})(z+1)} dz + B \\ &= A \int \frac{1}{\sqrt{z-1}} - \frac{i}{\sqrt{2}} \left[\frac{1}{\sqrt{z-1}} \left(\frac{1}{\sqrt{z-1} + i\sqrt{2}} - \frac{1}{\sqrt{z-1} - i\sqrt{2}} \right) \right] dz + B \\ &= 2A \left(\sqrt{z-1} - \frac{i}{\sqrt{2}} \text{Log}(\sqrt{z-1} + i\sqrt{2}) + \frac{i}{\sqrt{2}} \text{Log}(\sqrt{z-1} - i\sqrt{2}) \right) + B. \end{aligned}$$

The fact that $f(1) = 0$ implies $2A \left(\frac{\pi}{\sqrt{2}} \right) + B = 0$. First, we will find A . Notice that $\text{Arg}(A) = -\frac{\pi}{2}$ as that is the measure of the exterior angle of the final line segment. We conclude that $A = -i|A|$. Substituting this into the above equation yields $B = \sqrt{2}i|A|\pi$. Finding A and B now reduces to finding $|A|$.

Notice that $\text{Im}(f(x)) = 1$ for $x < -1$. Consider $x = -2$ and notice that

$$f(-2) = -2|A| \left(-\sqrt{3} + \frac{1}{\sqrt{2}} \ln|(\sqrt{3} + \sqrt{2})i| - \frac{1}{\sqrt{2}} \ln|(\sqrt{3} - \sqrt{2})i| \right) + \sqrt{2}i|A|\pi,$$

so $\text{Im}(f(-2)) = \sqrt{2}|A|\pi$. But, $\text{Im}(f(-2)) = 1$. We have that $\sqrt{2}|A|\pi = 1$ and so $|A| = \frac{1}{\sqrt{2}\pi}$. Thus, $A = \frac{-i}{\sqrt{2}\pi} = \frac{-i\sqrt{2}}{2\pi}$ and $B = i$. Substituting this into f yields

$$f(z) = \frac{-i\sqrt{2}}{\pi} \left(\sqrt{z-1} - \frac{i}{\sqrt{2}} \text{Log}(\sqrt{z-1} + i\sqrt{2}) + \frac{i}{\sqrt{2}} \text{Log}(\sqrt{z-1} - i\sqrt{2}) \right) + i.$$

Example 3.6 (Channel width formula). Observe that we can write the logarithm as a Schwarz-Christoffel transformation by letting $\text{Log}(z) = \int \frac{dz}{z}$ where there is only one vertex at $x_1 = 0$ having $\theta_1 = \pi$ and $w_1 = \infty$. Choosing $\theta_1 = \pi$ forces a semi-infinite

channel in the left half-plane. Here we will also have an infinite channel in the right half-plane, as shown in Figure ???. The channel width is $|s|$.

For points x^* on the real axis near $x_1 = 0$, we have

$$f(x^*) - f(-x^*) = -i\pi,$$

so the channel width is π .

First, we consider a Schwarz-Christoffel transformation where $\theta_{j_0} = \pi$ and $\theta_j \leq \pi$ for all $j \neq j_0$. By Equation ??, we have

$$f(z) = A \int \frac{dz}{(z - x_1)^{\frac{\theta_1}{\pi}} \cdots (z - x_{j_0}) \cdots (z - x_{n-1})^{\frac{\theta_{n-1}}{\pi}}} + B.$$

We will now show that $\lim_{z \rightarrow x_{j_0}} |f(z)| = \infty$ as $z \rightarrow x_{j_0}$ from the upper half-plane.

Notice that

$$f'(z) = \frac{1}{(z - x_1)^{\frac{\theta_1}{\pi}} \cdots (z - x_{j_0}) \cdots (z - x_{n-1})^{\frac{\theta_{n-1}}{\pi}}}.$$

Let $f'(z) = \frac{g(z)}{z - x_{j_0}}$ where $g(z) = (z - x_1)^{-\frac{\theta_1}{\pi}} \cdots (z - x_{n-1})^{-\frac{\theta_{n-1}}{\pi}}$. Here $g(z)$ is analytic near x_{j_0} and has a Taylor Series expansion about x_{j_0} . Then,

$$f'(z) = \frac{a_0}{z - x_{j_0}} + a_1 + a_2(z - x_{j_0}) + a_3(z - x_{j_0})^2 + \cdots$$

This implies that $f(z) = \text{Log}_{-\frac{\pi}{2}}(z - x_{j_0}) + h(z)$ where $h(z)$ is analytic. Moreover, $f(z) = \ln|z - x_{j_0}| + i\text{Arg}_{-\frac{\pi}{2}}(z - x_{j_0}) + h(z)$. We see that $\lim_{z \rightarrow x_{j_0}} |f(z)| = \infty$ because $\lim_{z \rightarrow x_{j_0}} \ln|z - x_{j_0}| = \infty$.

Next, we'll define the channel separation complex number s by

$$s = \lim_{r \downarrow 0} (f(x_{j_0} + r) - f(x_{j_0} - r)).$$

We see s labeled in Figure ?? and $|s|$ will give us the channel width. Observe that we can parameterize the upper-half semi-circle from $x_{j_0} - r$ to $x_{j_0} + r$ as $z(t) = x_{j_0} + re^{i(\pi-t)}$ for $0 \leq t \leq \pi$. Suppose that $s = \lim_{r \downarrow 0} (f(x_{j_0} + r) - f(x_{j_0} - r))$. Notice that

$$\begin{aligned} s &= \lim_{r \downarrow 0} [f(x_{j_0} + r) - f(x_{j_0} - r)] \\ &= \lim_{r \downarrow 0} [f(z(\pi)) - f(z(0))] \\ &= \lim_{r \downarrow 0} f(z(t)) \Big|_0^\pi \\ &= \lim_{r \downarrow 0} \int_0^\pi f'(z(t)) z'(t) dt \\ &= A \lim_{r \downarrow 0} \int_0^\pi \frac{z'(t) dt}{(z(t) - x_1)^{\frac{\theta_1}{\pi}} \cdots (z(t) - x_{j_0}) \cdots (z(t) - x_{n-1})^{\frac{\theta_{n-1}}{\pi}}} \\ &= A \lim_{r \downarrow 0} \int_0^\pi \frac{-ire^{i(\pi-t)}}{(x_{j_0} + re^{i(\pi-t)} - x_1)^{\frac{\theta_1}{\pi}} \cdots (re^{i(\pi-t)}) \cdots (x_{j_0} + re^{i(\pi-t)} - x_{n-1})^{\frac{\theta_{n-1}}{\pi}}} \\ &= A \int_0^\pi \lim_{r \downarrow 0} \frac{-i}{(x_{j_0} + re^{i(\pi-t)} - x_1)^{\frac{\theta_1}{\pi}} \cdots (x_{j_0} + re^{i(\pi-t)} - x_{n-1})^{\frac{\theta_{n-1}}{\pi}}} \\ &= A \int_0^\pi \frac{-i}{\prod_{j=1, j \neq j_0}^{n-1} (x_{j_0} - x_j)^{\frac{\theta_j}{\pi}}} \\ &= \frac{-A\pi}{\prod_{j=1, j \neq j_0}^{n-1} (x_{j_0} - x_j)^{\frac{\theta_j}{\pi}}}. \end{aligned}$$

Therefore, we see that

$$|s| = \frac{|A|\pi}{\prod_{j=1, j \neq j_0}^{n-1} |x_{j_0} - x_j|^{\frac{\theta_j}{\pi}}}.$$

We conclude with a brief remark of the importance of these results. When studying fluid flow, we may be interested in constructing the flow of an ideal fluid within

a polygonal domain P . It turns out that a Schwarz-Christoffel transformation will be useful in solving this problem, which is outlined in Section 7.3-Section 7.5 in [?]. The Schwarz-Christoffel transformation can also be instrumental in the determination of streamlines and isotherms. Several examples can be found in Section 6.4 in [?]. One can also use a Schwarz-Christoffel transformation in the calculation of air gap reluctance and inductance for magnetic circuits [?]. Furthermore, researchers have investigated using these particular transformations to produce grid generation algorithms for global ocean general circulation models [?]. We see that these transformations are used mostly in physics and engineering, but could be useful in other fields as well.

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